

CompositionMeasure for Approximate<ZeroConcentratedDivergence>

Michael Shoemate (@Shoeboxam)

This proof resides in “**contrib**” because it has not completed the vetting process.

Proves soundness of the implementation of **CompositionMeasure** for **Approximate<ZeroConcentratedDivergence>** in **mod.rs** at **commit f5bb719** (outdated¹).

1 Hoare Triple

Precondition

Compiler-verified

Types matching pseudocode.

Caller-verified

None

Pseudocode

```
1 class CompositionMeasure(ApproximateZeroConcentratedDivergence):
2     def composability( #
3         self, adaptivity: Adaptivity
4     ) -> Composability:
5         return Composability.Sequential
6
7     def compose(self, d_mids: Vec[Self_Distance]) -> Self_Distance:
8         rho_g, del_g = 0.0, 0.0
9         for rho_i, del_i in d_mids:
10             rho_g = rho_g.inf_add(rho_i)
11             del_g = del_g.inf_add(del_i)
12         return rho_g, del_g
```

Postcondition

Theorem 1.1. `composability` returns `Ok(out)` if the composition of a vector of privacy parameters `d_mids` is bounded above by `self.compose(d_mids)` under `adaptivity` `adaptivity` and `out-composability`. Otherwise returns an error.

Proof. By the postcondition of **InfAdd** we have that $\sum_i d_mids_i \leq \text{compose}(d_mids)$, where the summation is applied independently to rhos and deltas, and the comparison applies to both the global rho and global delta.

¹See new changes with `git diff f5bb719..32e0167 rust/src/combinators/sequential_composition/mod.rs`

Adaptivity	Sequential	Concurrent
Non-Adaptive	Lemma 8.2[BS16]	None
Adaptive	Lemma 8.2[BS16]	None
Fully-Adaptive	Theorem 1[WRRW23]	None

This table is reflected in the implementation of `composability` on line 2.

□

References

- [BS16] Mark Bun and Thomas Steinke. Concentrated differential privacy: Simplifications, extensions, and lower bounds, 2016.
- [WRRW23] Justin Whitehouse, Aaditya Ramdas, Ryan Rogers, and Zhiwei Steven Wu. Fully adaptive composition in differential privacy, 2023.